

The Wearables Museum

New formats (glTF and USDZ), new thumbnails, new previews, upgraded metadata and storage hosting to arweave. Durable and interoperable.

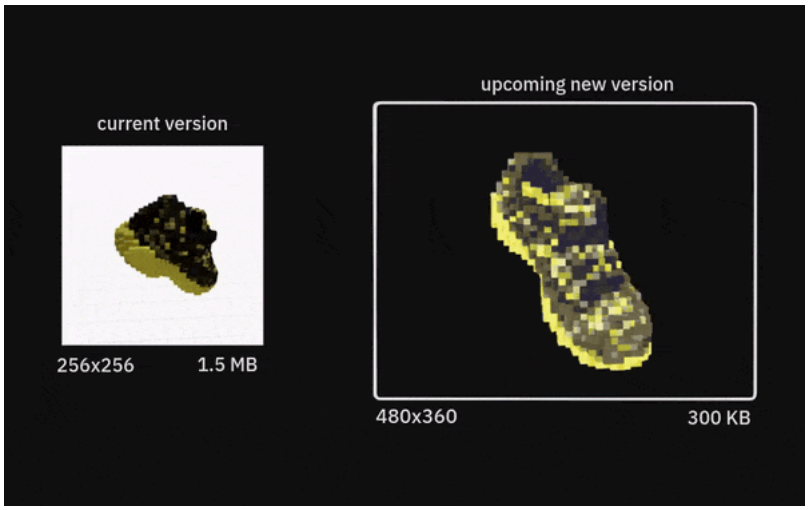
Quick summary of wearables 2.0:

- Upgraded file formats
  - glb with encoded xmp metadata in addition to vox
- Upgraded to decentralized storage
  - From voxels.com to arweave for all assets
- Upgraded thumbnails
  - Transparent gifs, ~9x smaller file size + 50% bigger resolution
- Upgraded previews
  - HTML files hosted on arweave with AR+VR previews
- Upgraded metadata
  - Updated the links, added fields like glb + collection info



New thumbnails

50% bigger resolution, 5x smaller filesize, transparent background, still GIF format.



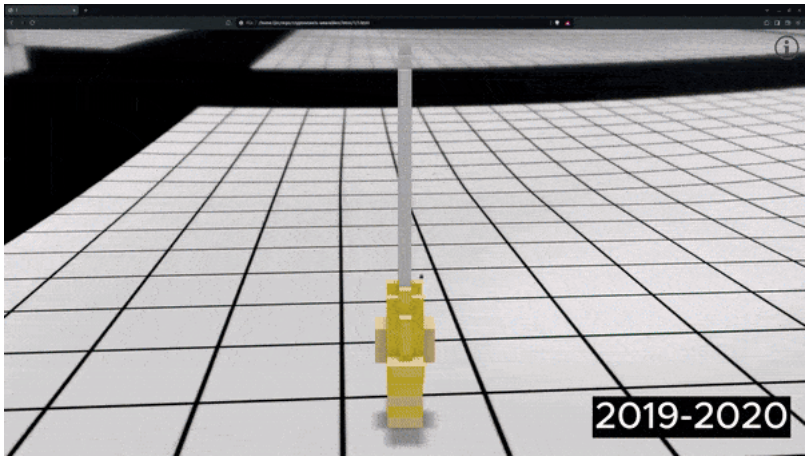
The thumbnails for all the wearables have been completely remade

New previews

Different skyboxes to represent the years a wearables collection was minted. Each skybox is a 360 equirectangular picture from a VRChat world that Voxels content was ported into and reflects the era:

- 2019-2020: Black and white era
- 2021: Color emerges, calm before the storm
- 2022: NFT and metaverse hype cycle peaks
- 2023: Desert to represent the winter that followed
- 2024: Half voxels half sea to represent WIP year + islands

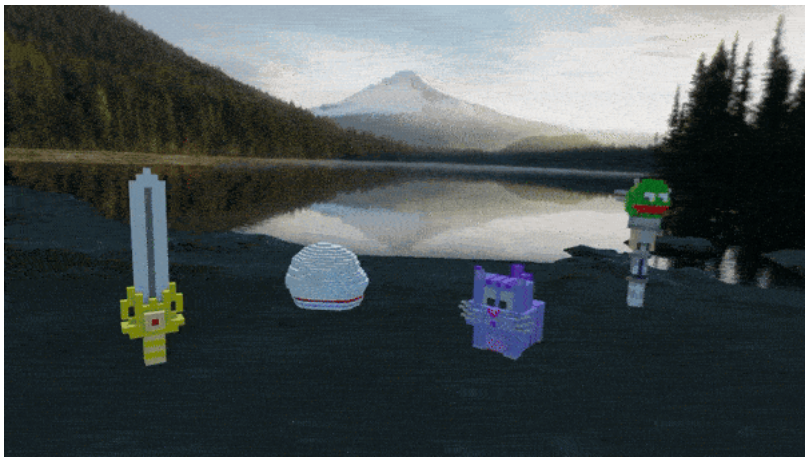
View a demo: https://arweave.net/DQzDSe2GFKopqmFOuwdeflWA53dtt6wy-6k609QiMg4



Different skyboxes per year the collection was minted

AR + VR Support via glTF and USDZ

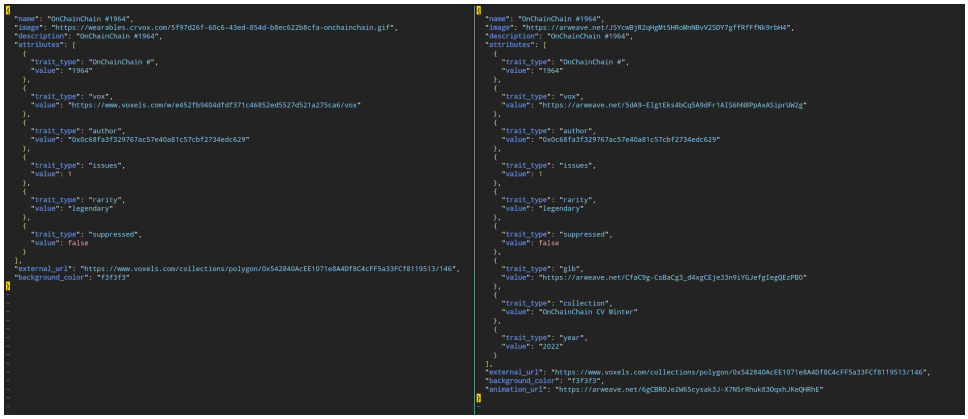
Every wearable can now be previewed in AR and VR. Tested across android and IOS devices including the Apple Vision Pro.



You can view the wearables via Safari on ios and Apple Vision Pro

New metadata

The NFT metadata for every wearable across every collection in the snapshot has been updated. On the right is a preview of the new metadata with arweave links and new metadata fields such as glb, collection, year, and animation_url for the HTML preview.



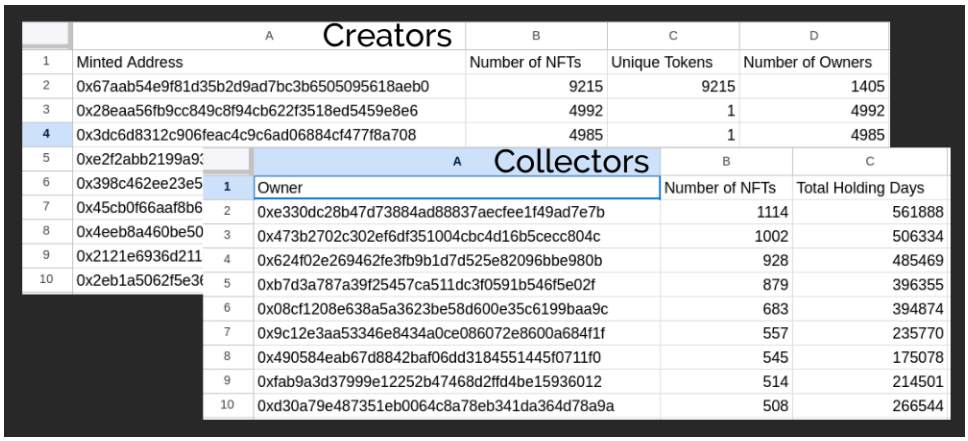
On the left is the old metadata, on the right is the new metadata with arweave links, glb link, collection metadata, and a link to the HTML preview

Mission complete!

Blockchain Snapshot and Stewardship

Date taken: **March 12, 2024**

- **https://docs.google.com/spreadsheets/d/19Fab72iK5Ks9KeI_ArY79m0wxyL10s45obZr1FGnKKo/edit?usp=sharing**
    - https://www.voxels.com/collections



Creators

	Minted Address	Number of NFTs	Unique Tokens	Number of Owners
2	0x67aab54e9f81d35b2d9ad7bc3b6505095618aeb0	9215	9215	1405
3	0x28eaa56fb9cc849c8f94cb622f3518ed5459e8e6	4992	1	4992
4	0x3dc6d8312c906feac4c9c6ad06884cf477f8a708	4985	1	4985
5	0xe2f2abb2199a93			
6	0x398c462ee23e5			
7	0x45cb0f66aaf8b6			
8	0x4eeb8a460be50			
9	0x2121e6936d211			
10	0x2eb1a5062f5e36			

Collectors

	Owner	Number of NFTs	Total Holding Days
2	0xe330dc28b47d73884ad88837aecfee1f49ad7e7b	1114	561888
3	0x473b2702c302ef6df351004cbc4d16b5cecc804c	1002	506334
4	0x624f02e269462fe3fb9b1d7d525e82096bbe980b	928	485469
5	0xb7d3a787a39f25457ca511dc3f0591b546f5e02f	879	396355
6	0x08cf1208e638a5a3623be58d600e35c6199baa9c	683	394874
7	0x9c12e3aa53346e8434a0ce086072e8600a684f1f	557	235770
8	0x490584eab67d8842baf06dd3184551445f0711f0	545	175078
9	0xfab9a3d37999e12252b47468d2ffd4be15936012	514	214501
10	0xd30a79e487351eb0064c8a78eb341da364d78a9a	508	266544

The Cryptovoxels community has designed over **34,000** unique wearables to **111,557 collectors** holding a total of 350,263 NFTs!

For sake of comparison, Decentraland has about **6840** unique wearables designs (Dune sources say ~10k) made by ~1587 creators and have about **4531 collectors**.

Number of Cryptovoxels wearables minted per year:

- 2020: 3138
- 2021: 21792
- 2022: 8621
- 2023: 438
- 2024: 26 (as of snapshot date)

Points

Each voxels parcel owned is 1 point = 7,937 Wearbles creator impact (see below) = 11,867 Total = 19804

I asked chatgpt to help me come up with an algorithm for distributing voting power amongst wearables creators:

> To create a fair distribution of points based on the impact of minted NFTs, considering factors such as the number of NFTs minted, the number of unique tokens, and the number of owners, you can use a weighted scoring system. Here's a suggested algorithm:
>
> Normalize the data: Normalize each of the three metrics (Number of NFTs, Unique Tokens, Number of Owners) to a common scale between 0 and 1. You can use min-max normalization for this purpose.
>
> Assign weights: Assign weights to each normalized metric based on their importance in determining impact. These weights can be subjective and depend on your specific use case. For example, you may decide that the number of NFTs minted is more important than the number of owners.
>
> Calculate scores: Calculate a composite score for each address by multiplying each normalized metric by its corresponding weight and summing them up.
>
> Distribute points: Allocate points to each address based on their composite score. You can distribute points proportionally or use a different method based on your preference.

We then round the points and add +1 for sake of being a wearables creator. For transparency sake, this is the python code that was used to calculate the results:

```
import pandas as pd
from sklearn.preprocessing import MinMaxScaler

# Read data from CSV file
df = pd.read_csv('creators.csv')

# Normalize the data
scaler = MinMaxScaler()
df[['Number of NFTs', 'Unique Tokens', 'Number of Owners']] = scaler.fit_transform(df[['Number of NFTs', 'Unique Tokens',

# Define weights
weights = {
    'Number of NFTs': 0.3,
    'Unique Tokens': 0.2,
    'Number of Owners': 0.5
}

# Calculate composite score
df['Composite Score'] = (df['Number of NFTs'] * weights['Number of NFTs']) + (df['Unique Tokens'] * weights['Unique Tokens

# Distribute points based on composite score
total_points = 10000  # Total points to distribute
df['Points'] = (df['Composite Score'] / df['Composite Score'].sum()) * total_points

# Round the points and convert to integers
df['Points'] = df['Points'].round().astype(int)

# Add 1 point to each address
df['Points'] += 1

# Export results to CSV
df[['Minted Address', 'Points']].to_csv('results.csv', index=False)

print("Results exported to 'results.csv'.")
```

Snapshot strategies

In addition to a standard strategy for who owns a Voxels parcel, we implemented the whitelist weighted strategy that incorporated the points calculated in the steps above for wearables creators to have a strong voice in voting.

```
whitelist weighted
{
  "symbol": "ABC",
  "addresses": {
    "0xa478c2975ab1ea89e8196811f51a7b7ade33eb11": 5,
    "0xeF8305E140ac520225DAf050e2f71d5fBcC543e7": 2
  }
}
```

Distribution Strategies

What is the next step? Here's a couple ideas I got:

1. Create a guide for creators to update collections themselves

The creator of a collection **overwrites** the existing NFT metadata with the new metadata, thereby upgrading the NFTs everyone already has to the new decentralized versions.



This strategy is like overwriting the old NFTs by writing to the contract

Benefits:

- no remint, just update existing contracts
  - 1 time transaction vs airdropping
  - maintain provenance, collectors already have it
- unsold copies can find new market value as durable and interoperable wearables

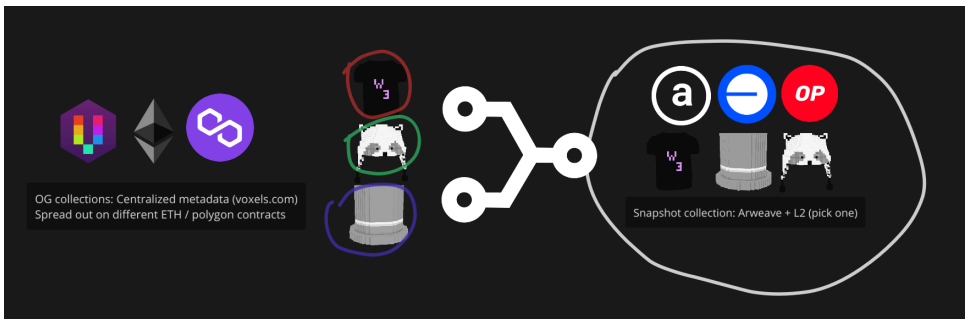
Downsides:

- Will it break things in voxels.com? Need to test
  - can still have `vox` as trait_type
  - could create a baseURI like `https://www.cryptovoxels.com/c/7/{id}` with Arweave for linking to the metadata
- Would still be hard to update all collections
- Could look messy if mix of old + new wearables

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2. Create a new wearable collection on a L2

Similar to a fork. It would not be an upgrade to the Voxels wearables, but rather a new collection to immortalize a ***SNAPSHOT*** of all the wearables. This option is like having an on-chain mirror.



Left to right = old to new. We can consolidate all the collections to 1 for the snapshot, and a DAO to steward it

Benefits:

- Full upgrade: All wearables 2.0 would look really nice together
- Immortalize the whole snapshot of wearables as a NFT collection

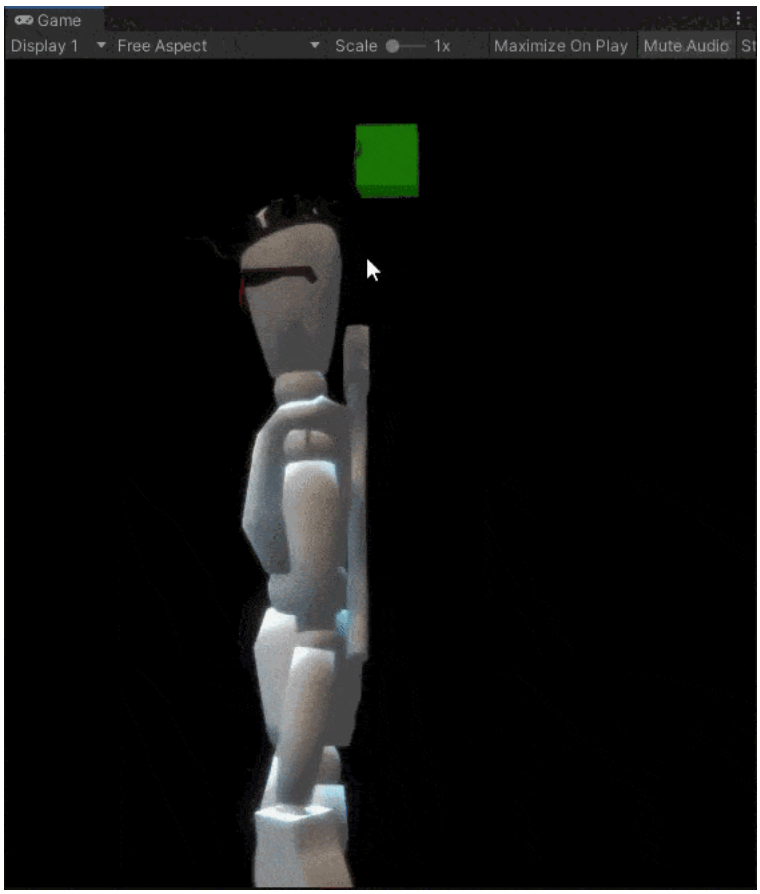
- Can airdrop or allow list previous owners to mint or burn redeem?
- Unsold copies can find new market value as durable and interoperable wearables
    - Use a splits contract to pay creators from sales?

Downsides:

- It might be confusing when searching for a wearable to see 2 similar results
- 100k+ collectors, many might not know + still expensive to airdrop
- It turns 800 creator owned collections into 1 collection
    - Perhaps this can be seen as a good thing for discoverability!

3. Create a NFT Avatar Collection

We mint all costumes as VRM avatar NFTs and create a pool of shared assets to generate into new avatars. The mannequin base mesh can be an OG trait, but we don't have to be limited to it.



Old gif of boom tools to represent creating an NFT avatar collection for Voxels

Benefits

- By minting avatars, not wearables, we can reduce some confusion
    - No duplicate wearables in marketplace
- Can broaden the appeal of Voxels to new people
- We can make the collection look absolutely stunning
- Feels fun and innovative, like metaverse passports + immigration cards
- This option can potentially pair with another option

- Can use splits contract to drive sales to a DAO

Downsides:

- There's a lot of wearables, not all of them might make it in
- What would be a fair split?
    - Maybe based on number of factors: wearables contributed, support for openvoxels, etc
- Might be hard to get permission to use wearables from a bunch of creators?

Costume -> VRM Avatar Export

For sake of posterity, since a whole post could be dedicated to this topic, here is a preview of Voxels costumes to VRM export. VRM avatars from costumes have been made and hand tailored for finishing touches for all Openvoxels supporters. We also exported VRM avatars for all parcel holders. What we do next is TBD :)



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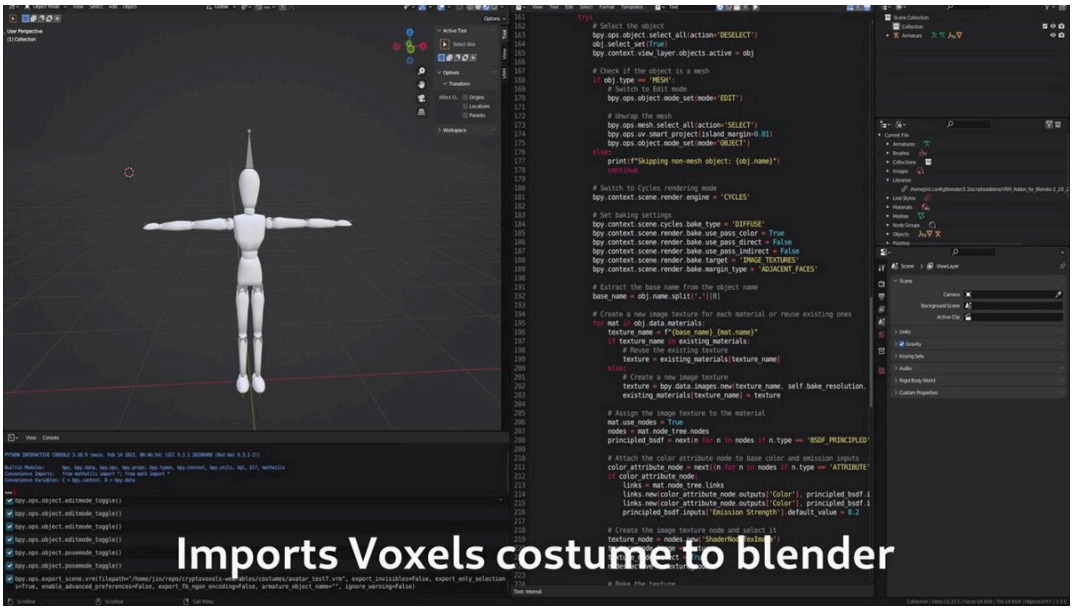
@dankvr





Citizens of @cryptovoxels, I've achieved VRM export

Takes in a costume JSON, converts the skin to texture, bakes the glTF wearables, and exports a working VRM #screenshotsaturday #blender @openvoxels





115

2:38 AM • Apr 7, 2024

Support Openvoxels

Donate to our juicebox, get $OPENVOX tokens, which can be used for minting exclusive swag. I think we will also give Openvoxels supporters voting power in our snapshot in the future. After shipping wearables and VRM avatar exports, there's a few exciting paths we can take on what to focus on next that could use your input!

Juicebox | Fund Your Thing

Juicebox is the programmable funding platform for crypto and web3. Fund, operate, and scale your project transparently. Community DAO owned, on Ethereum.

https://juicebox.money



Last note: I think we will be doing an art contest for wearables soon, stay tunes and follow Openvoxels on X for more info!

